Data-Driven Inventory Optimization

Objective & Success Metrics

Goal

Leverage data-driven demand forecasting and replenishment strategies to reduce stockouts and lost sales while maintaining optimal inventory levels across the retail network.

Objective

Improve product availability (Fill Rate) while ensuring consistent and efficient inventory control across all stores and SKUs.

Business Challenges

- **Stockouts** → Revenue loss and reduced customer satisfaction
- **Overstock** → Capital tied up in inventory and higher holding costs

Key Performance Indicator (KPI)

Fill Rate (Service Level)

→ Percentage of customer demand fulfilled without stockouts

Supporting Metrics

- **Forecast Accuracy:** WAPE, MAPE, Forecast Bias
- **Availability:** Stockout Rate, Lost Sales (Units / ₹)
- **Inventory Management:** Days of Holding (DOH), Overstock Rate, Inventory Turns

Data & Methodology (ETL + Forecasting)

Data Coverage (Store–SKU–Day Level)

- **Sales Data**: Units sold and sales revenue
- **Inventory Data**: Daily on-hand inventory snapshots
- **Purchase Orders**: Order quantities and realized lead times
- **Product Master Data**: Category, cost, shelf life, minimum order quantity (MOQ)
- **Calendar Attributes**: Day of week, promotion indicator, holiday indicator

ETL Methodology

- Construct a complete daily time series for each store–SKU combination
- Standardize identifiers, address missing dates, and remove duplicate records
- Detect and treat demand outliers using the IQR method
- Forward-fill inventory values to maintain continuity for planning and analysis

Forecasting Methodology

- **Granularity**: Store–SKU–Day
- **Forecast Horizon**: 28 days (4 weeks)
- **Models Evaluated**:
  - Seasonal Naive model (weekly baseline)
  - Regression model *(selected)*
- **Features Used**: Day of week, promotion flag, holiday flag
- **Model Selection Criteria**: Lower WAPE, reduced forecast bias, and improved promotion responsiveness

Data & Methodology (Replenishment Policy)

Replenishment Policy Type

Store–SKU–level **Order-Up-To (Base Stock) Replenishment Policy**

Key Inputs

- Average daily demand (based on the most recent 8 weeks)
- Demand variability (standard deviation of daily demand)
- Supplier lead time (median value, with a default fallback where unavailable)
- Category-specific service level targets

Core Calculations

- **Safety Stock** = $z \times \sigma \times \sqrt{(\text{Lead Time})}$
- **Reorder Point (ROP)** = (Average Daily Demand × Lead Time) + Safety Stock
- **Order Quantity** = Quantity required to raise inventory to **30 days of coverage**

Operational Constraints Applied

- Minimum Order Quantity (MOQ) requirements
- Shelf-life constraints for perishable SKUs
- Optional store-level capacity limits

Key Insights

- Stockouts are **highly concentrated**, rather than evenly distributed across the network
- A **small set of categories** accounts for the majority of stockout days
- Categories with frequent stockouts exhibit **higher demand volatility**
- **Promotions significantly increase demand**, creating under-forecasting risk when not explicitly modeled
- A **limited number of stores** contribute disproportionately to overall stockout incidence
- **Overstock risk is primarily driven by excessive Days of Holding (DOH > 45 days)**
- **Targeted, category- and store-level replenishment interventions** can reduce stockouts without increasing overall inventory levels

Forecast Performance & Chosen Approach

Evaluation Metrics

- **WAPE (Primary Metric)**: Volume-weighted measure of forecast accuracy
- **MAPE (Diagnostic Metric)**: Interpretability check, excluding zero-demand days
- **Forecast Bias**: Indicator of systematic over- or under-forecasting risk

Performance Results

- The **regression model consistently outperforms the Seasonal Naive baseline** on WAPE
- Forecast bias is **closer to zero**, lowering the risk of both overstock and stockouts
- Performance remains **stable across categories**, including high-volume SKUs

Rationale for Model Selection

- Effectively captures **weekly seasonality** and **promotion/holiday effects**
- Produces a **more stable demand signal** for replenishment decisions
- Reduced bias minimizes the risk of **inventory imbalance**

Validation Diagnostics

- Strong alignment between **actual and forecast demand trends**
- Forecast error distribution **centered around zero**, indicating unbiased predictions
- **Category-level WAPE comparisons** used to validate robustness across segments

Replenishment Plan: How It Works

Replenishment Policy

Store–SKU–level **Order-Up-To (Base Stock) Replenishment Policy**

How the Replenishment Plan Works

- Estimate demand using the most recent **8-week average** and associated variability
- Incorporate **supply uncertainty** through realized supplier lead times
- Safeguard target service levels using **category-specific safety stock**
- Initiate replenishment when **on-hand inventory falls below the Reorder Point (ROP)**
- Place orders to restore inventory to a **target level equivalent to 30 days of coverage**

Key Inputs

- Average daily demand (last 8 weeks)
- Demand variability (standard deviation)
- Supplier lead time
- Category-level service level targets

Core Calculations

- **Safety Stock** = $z \times \sigma \times \sqrt{(\text{Lead Time})}$
- **Reorder Point (ROP)** = (Average Daily Demand × Lead Time) + Safety Stock
- **Order Quantity** = Quantity required to raise inventory to **30 days of cover**

Replenishment Plan: Examples

Example 1 — Stockout-Risk SKU

- **Average Daily Demand**: 8 units
- **Supplier Lead Time**: 4 days
- **Service Level Target**: 97%
- **Current Inventory Status**: On-hand inventory below the Reorder Point

Action

Initiate replenishment to restore inventory to a **30-day coverage level**, preventing an imminent stockout and avoiding potential lost sales.

Example 2 — Overstock-Risk SKU

- **Days of Inventory on Hand (DOH)**: Greater than 45 days
- **Inventory Position**: Demand coverage significantly exceeds near-term requirements

Action

- Reduce or temporarily pause future replenishment orders
- Rebalance inventory by transferring excess stock to higher-demand locations
- Apply targeted discounts where appropriate to accelerate sell-through

30-Day Impact Estimate (Assumptions & Scenarios)

Assumptions & Methodology

- A **stockout** is defined when on-hand inventory falls below the **Reorder Point (ROP)**
- **Lost sales** are estimated using a proxy: *Average Daily Demand × Supplier Lead Time*
- Business impact is assessed over a **30-day forward-looking horizon**
- Execution uncertainty is modeled using a **±20% variance** around expected outcomes

Estimated Business Impact

- **Meaningful reduction in stockout days** for high-risk store–SKU combinations
- **Material recovery of lost sales (₹)** driven by improved product availability
- **Controlled increase in inventory value**, reflecting an intentional trade-off to achieve higher service levels

Scenarios Evaluated

- **Base Case**: Expected execution performance
- **Best Case (+20%)**: Strong replenishment execution and operational alignment
- **Worst Case (–20%)**: Partial execution due to operational constraints or friction

Key Risks & Recommendations

Risks, Recommendations & Expected Outcomes

Key Risks & Limitations

- Linear forecasting may not fully capture **non-linear demand patterns**
- Promotional uplift assumed **uniform across stores and SKUs**
- Lead-time variability **partially captured** using historical medians
- Lost sales estimated using a **proxy**, not actual unmet demand

Current Mitigations

- **Conservative, category-specific service levels**
- **Scenario-based impact assessment** (Base / Best / Worst)
- Continuous **category- and store-level monitoring**

Recommendation

- **Phased rollout** of the replenishment policy, starting with:
  - **Top stockout-driving categories**
  - **High-impact store–SKU combinations**
- Enables **rapid value capture** while limiting operational risk